europass



Heeta Bhanushali

Passport: Z7522232 | **Date of birth:** 29/04/2004 | **Nationality:** Indian | **Gender:** Female |

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Address: 400060, Mumbai, India (Home)

ABOUT ME

Final-year Computer Engineering student passionate about low-level systems, language design, and security. Driven by clean engineering, strong fundamentals, and problems that demand both theoretical depth and practical skill.

WORK EXPERIENCE

SOFTWARE ENGINEER INTERN – TRANSLINE AIR CARGO SERVICES PVT. LTD. – 06/2024 – 11/2024 – MUMBAI, INDIA

- Designed relational database schema replacing Excel-based operations across shipment tracking, custom documentation, invoicing, and client management
- Built ETL pipelines to migrate and clean historical data from unstructured spreadsheets
- Enforced business rules through database-level constraints and developed automated reporting

EDUCATION AND TRAINING

2022 – CURRENT Mumbai, India
BACHELORS IN TECHNOLOGY IN COMPUTER ENGINEERING Dwarkadas J. Sanghvi College of Engineering

Website https://www.djsce.ac.in/ | **Final grade** 8.94 (average of 7 semesters) | **Level in EQF** EQF level 6

2020 – 2022 Mumbai, India
HIGHER SECONDARY CERTIFICATE Mithibai College

Website https://mithibai.ac.in/ | **Field of study** Field unknown | **Level in EQF** EQF level 4

2020 Mumbai, India
SECONDARY SCHOOL CERTIFICATE Smt. Surajba Vidya Mandir

Website https://surajbavidyamandir.org/

PROJECTS

12/2025 – 02/2026
TransitCore — Formally-Grounded Workflow Engine

- Built a workflow engine enforcing declarative state transitions with a custom expression language and EBNF grammar
- Implemented static verifier proving temporal logic properties with atomic transitions and automatic rollback
- Developed REST API with optimistic locking, audit trails, auto-generated diagrams, and 93% test coverage

Technologies: Python, Lark (EBNF parser generator), SQLAlchemy, Graphviz, Hypothesis

Link https://github.com/heetabhanushali/TransitCore.git

02/2025 – 05/2025
MiniLang Compiler & Static Analyzer

- Built a complete compiler with lexer, parser, type checker, optimizer, and C code generator
- Implemented four optimization passes (constant folding, dead code elimination, propagation, strength reduction)
- Designed static analysis engine with complexity metrics and deployed WebAssembly browser playground with Monaco Editor

Technologies: Rust, WebAssembly, wasm-bindgen, Logos, miette, serde, Monaco Editor, Vercel

Links https://github.com/heetabhanushali/minilang_compiler.git | https://minilang-playground.vercel.app/

12/2024 – 01/2025

Gloves Shop — E-Commerce Microservices Platform

- Built 8 polyglot microservices across 12 containerized services with database persistence and async messaging
- Implemented CI/CD pipelines with automated builds, image publishing, and infrastructure-as-code for cloud deployment
- Configured real-time monitoring and alerting dashboards with container-level metrics

Technologies: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Prometheus, Grafana, Node.js, Python, Java, Go, PHP, MongoDB, MySQL, Redis, RabbitMQ

Link https://github.com/heetabhanushali/gloves-shop-cicd.git

09/2024 – 11/2024

Cross-Platform Packet Sniffer & Network Security Monitor

- Built a real-time network analyzer with packet inspection across 13 protocols (Layers 2–7) and heuristic-based threat detection
- Designed multi-threaded capture engine with concurrent parsing, statistics aggregation, and cross-platform support
- Developed interactive dashboard with live packet streaming, protocol charts, and security alerts

Technologies: Python, Scapy, Flask, Multithreading, REST APIs, JavaScript, Chart.js, HTML/CSS

Links https://github.com/heetabhanushali/packet-sniffer.git | https://packet-sniffer-bpub.onrender.com | https://drive.google.com/file/d/1O33FlX99Hey1hdfA9x1azEwLy3Ts-oXn/view?usp=sharing

12/2023 – 05/2024

HelloChat – Real-time Social Chat Application

- Built an offline-first encrypted chat app with end-to-end encryption and cross-device key recovery
- Engineered offline message queue with exponential backoff, deduplication, and bidirectional sync
- Developed real-time messaging, typing indicators, read receipts, soft-delete conversations, and AI chatbot integration

Technologies: Flutter, Dart, Supabase, PostgreSQL, Hive, AES-256-GCM, WebSocket

Link https://github.com/heetabhanushali/flutter-chat-app.git

PUBLICATIONS

2025

Bias Mitigation in Language Models using a Multi-Dimensional Evaluation and Comparison

Authors: Arham Shah, Jai Jain, Heeta Bhanushali, Granth Jain, Nilesh Patil, Chinmay Raut | **Journal Name**: ICRMLAI'25

2025

TADR: Typology-Aware Dynamic Routing for Zero-Shot Cross-Lingual Transfer

Authors: Heeta Bhanushali, Jai Jain, Ameya Suryawanshi, Meera Narvekar, Kranti Ghag, Khushali Deulkar | **Journal Name**: CSCT 2025

LANGUAGE SKILLS

Mother tongue(s): **KUTCHHI** | **GUJARATI**

Other language(s): **MARATHI** | **HINDI** | **ENGLISH** | **GERMAN**

SKILLS

Languages

C | C++ | Java | Python | Rust | Dart | JavaScript

Frameworks & Tools

FastAPI | Flask | Flutter | Firebase | Supabase | GitHub Actions | Docker | SQLAlchemy

Concepts

Compiler Design | Data Structures & Algorithms | REST APIs | CI/CD | WebAssembly | Encryption | Microservices | Static Analysis | Finite State Machines | Code Optimization